- *Service*—A Deployment (a set of Pods) can be exposed to other nodes in the cluster or outside by defining a Service that also takes care of balancing the load between Pod instances.

NOTE Throughout the book, I will write Kubernetes resources with capital letters to distinguish them from when the same terms are used with different meanings. For example, I'll use *service* when referring to an application, whereas I'll write *Service* when I mean the Kubernetes object.

When you want to run a new application, you can define a *resource manifest*, a file that describes the desired state for the application. For example, you might specify that it should be replicated five times and exposed to the outside world through port 8080. Resource manifests are usually written using YAML. You can then use the kubectl client to ask the control plane to create the resources described by the manifest. In the end, the control plane processes the request using its internal components and creates the resources in the worker nodes. The control plane still relies on a container registry to fetch the image defined in the resource manifest. The workflow, again, is shown in figure 2.10.

2.4.2 Running a Spring application on Kubernetes

Let's go back to the Polar Bookshop project. In the previous section, you containerized the Catalog Service application. Now it's time to deploy it to a cluster using Kubernetes. You already have a cluster up and running in your local environment. What you need is a resource manifest.

The standard way to interact with Kubernetes is through declarative instructions that you can define in a YAML or JSON file. I will show you how to write resource manifests in chapter 7. Until then, you can use the Kubernetes CLI as you did with Docker earlier.